

Stage 1 report

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1 Literature review

A substantial body of research in cognitive science and marketing has documented systematic deviations from the assumptions of classical rational choice models, emphasizing the importance of context effects in human decision-making. Foundational work identifies phenomena such as the similarity effect [Tversky \(1972\)](#), where introducing a new option disproportionately cannibalizes choices among similar items; the attraction (decoy) effect [Huber et al. \(1982\)](#), in which adding a dominated alternative increases the appeal of a target option; and the compromise effect [Simonson \(1989\)](#), where intermediate options become more attractive in the presence of extreme alternatives. These effects complement broader behavioral insights such as loss aversion and framing biases [Kahneman and Tversky \(1979\)](#); [Tversky and Kahneman \(1991\)](#), collectively revealing the limitations of standard Random Utility Models (RUM).

To address these limitations, a stream of context-dependent extensions of MNL has been developed, including Context MNL models [Ko and Li \(2024\)](#); [Yousefi Maragheh et al. \(2018\)](#); [Seshadri et al. \(2019\)](#), which allow utilities to depend linearly on other items in the choice set. However, these approaches typically ignore item-specific features and are limited to linear effects, reducing their applicability in richer environments.

More recent work incorporates deep learning architectures—using ResNet for assortment-dependent modeling [Wang et al. \(2023b\)](#) or Deep Set architectures [Pfannschmidt et al. \(2022\)](#); [Rosenfeld and Ravikumar \(2020\)](#) to handle permutation-invariant choice sets. Other studies leverage attention-based neural networks [Ko and Li \(2024\)](#); [Wang et al. \(2023a\)](#) to capture higher-order interactions, although they often rely on single-head attention for theoretical analysis. In contrast, recent advances highlight the expressive advantages of wide multi-head attention and its connections to neural tangent kernel theory.

A parallel line of research investigates consider-then-choose models [Akchen and Mitrofanov \(2024\)](#); [Aouad et al. \(2021\)](#); [Brady and Rehbeck \(2016\)](#); [Gallego and Iravani \(2024\)](#); [Wang \(2022\)](#), which posit that consumers first screen alternatives to form a consideration set before making a final choice—a framework that aligns closely with realistic multi-stage decision processes.

Together, these developments illustrate both the limitations of classical RUM and the growing interest in flexible, context-aware choice models.

2 Options of choice

We follow the instructions to use DeepHalo in stage 1, and performed tests on synthetic dataset. In the next stage, we would like to explore methods that considering unobserved market-product characteristics.

37 3 Answers

38 3.1 Are there any errors or unclear parts in the paper? If there are, please point them out 39 and explain the unclear parts in the report.

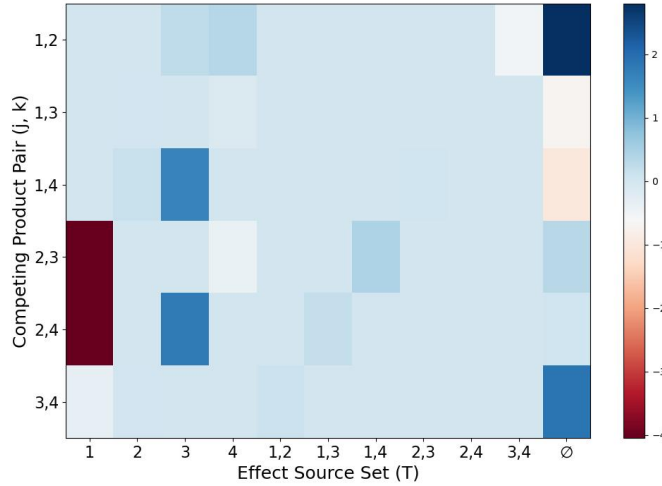
40 (a) In equation (3) the authors use $\mu_j(x_j, X_R)$ that takes two inputs, whereas in equation (2)
41 μ_j is defined to only take one input. It is naturally confusing here, and it should be like
42 $\mu_j(X_{R \cup \{j\}})$.

43 (b) normalizer $\frac{1}{S}$ across the paper, e.g., in equation (4), (6), (8), (9), should be $\frac{1}{|S|}$ since S is a
44 set.

45 (c) Situations/settings for using equation (5) or (8) are unclear. To me, a reasonable guidance
46 should be that, when higher-order context effects are involved, (8) should be used as it
47 allows faster growth in representational power with depth.

48 3.2 Do you get the same results as their synthetic data test?

49 Yes, after running our provided code `hypothetic.py`, we generate the visualization of the relative
50 Halo effects across beverages, presented below. The result aligns with figure 1 in Zhang(25).

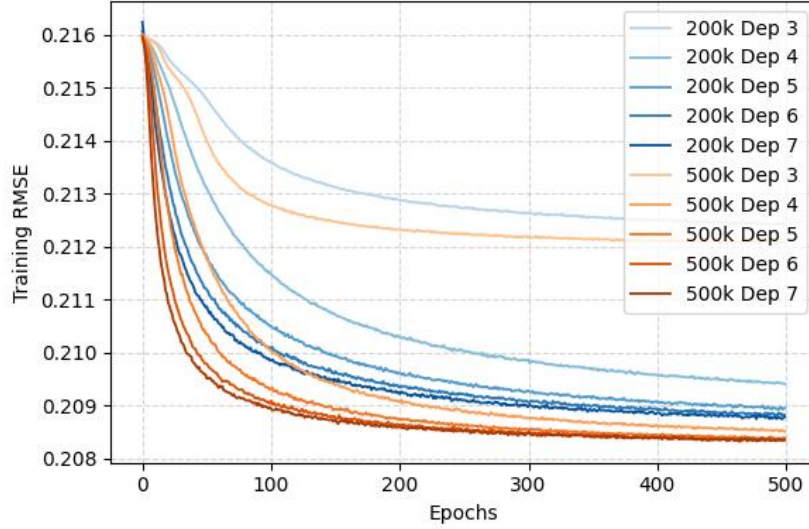


51 In addition, by executing the supplied script `synthetic.py`, we generate the comparisons of train-
52 ing loss curve among different depth of order, presented below. We follow the parameter setting in
53 table 4. The result aligns with Figure 3 in Zhang(25), though the RMSE values are a bit different.
54 Ours are in the range 0.208 to 0.216, and the range in Zhang(25) is 0.02 to 0.06. This is potentially
55 due to different random seed setting in generating synthetic dataset.

56 3.3 What additional tests could you come up with to verify the results?

57 (a) **To present computational performance statistics:** How fast/slow is it, compared with
58 the other DNNs like FateNet and TCNet? We need to know this because perfor-
59 mance/computation cost is an important trade-off in practical situations.

60 (b) **To further illustrate the utility of interaction order controlling:** It seems that in table
61 3, 7, and 8, the difference of Negative log-likelihood between DeepHalo and TCNet is not
62 statistically significant. According to Appendix C.2.2, the authors tune the hyperparameters
63 of DeepHalo based on validation performance and subsequently controls interaction order,
64 i.e., number of layers, in $\{4, 5\}$.



In LPMC and Expedia datasets, the low-order interactions might dominate, so that TCNet matches DeepHalo’s accuracy. It is interesting to see if DeepHalo’s interaction order controlling could be distinct from TCNet (and other transformer-based approaches) in some sythetic or real-wrold datasets where high-order interactions dominate.

- (c) **Q: To compare models in featureless datasets:** While many benchmark models are compared in datasets with features, e.g.,table 3, while in featureless datasets, e.g.,table 1, only a few models are tested. Models like FateNet and TCNet can also be applied to featureless cases.

3.4 Comparing to the reproducing kernel Hilbert Space Choice model of Yang(25), what are the pros and cons of Zhang(25)?

Pros

- (a) **Interaction order control:** Yang(25) can essentially be viewed as one of the transformer-based models, which implicitly allow interactions across all items, inevitably resulting in full-order interactions ($|S| - 1$). While Zhang(25) provides explicit layer-wise control, where each layer adds exactly one level of interaction. This control adds more regularization and tractability.
- (b) **Expressive structure:** Zhang(25) eliminates the need for query/key/value projections and softmax attention that exist in transformer. The attention mechanism is replaced by a linear transformation followed by an activation function. This results a linear, rather than quadratic, computational cost.
- (c) **Better scalability compared to fixed kernel:** Fixed kernel-based model (e.g., Gaussian, Matérn, Polynomial etc.) naturally leads to memory bottleneck when dataset size is large, though Random Feature and NTK in Yang(25) are introduced to address this challenge.

Cons

- (a) **Optimization challenge:** For fixed and random kernels in Yang(25), the problem becomes a finite-dimensional convex optimization over coefficients, which can be solved using convex optimization solvers. However, DeepHalo in Zhang(25) and NTK in Yang(25) requires non-convex training, sensitive to hyperparameters, initialization, and architecture.
- (b) **Overfit easily without sufficient samples:** Neural models overfit easily without sufficient samples. As can be seen from the better out-of-sample performance in fig 1.(b) and table 5, fixed kernels in Yang(25) possess better generalization capability.

96 **3.5 Do you think these models are suitable for the task of demand estimation of credit card**
97 **offers? What are the potential problems or challenges?**

98 I think, in principle, these models are well suited because credit-card offer choices involve:

- 99 • multiple competing alternatives (e.g., cashback, miles, fee-free, student, premium)
- 100 • context effects (card framing, order, competing offers)
- 101 • strong interactions between customer features and offer features
- 102 • non-compensatory behaviors (customers prioritize certain attributes heavily)

103 In an ideal situation, where each customer knows the full set of choices (including alternatives they
104 could have chosen), it is very similar to the beverage data set in Zhang(25). However, there are
105 several challenges under real-world conditions:

- 106 (a) **Data sparsity:** Most customers see only a few offers, not all the cards the market is pro-
107 viding. DeepHalo requires large, rich choice-set data, and banks often don't have it or can't
108 use it fully due to privacy. As a consequence, neural models overfit easily without sufficient
109 exposure-level variety.
- 110 (b) **Customer heterogeneity:** Credit card demand depends heavily on factors like income,
111 spending behavior, travel patterns, risk assessment etc. DeepHalo can model heterogeneity
112 but needs vast feature engineering and hence requires massive training data, which is non-
113 trivial.
- 114 (c) **Constrained assortment:** There are frequent constraints in credit card offers. For ex-
115 ample, a credit card account must be accompanied by a main account with some certain
116 information. salary requirement.

117 **3.6 What are other models, traditional non-neural network based, or ML models that you**
118 **think are more suitable or at least worth trying? Why?**

119 Several non-neural network models are highly suitable—and often preferable—for discrete choice
120 and demand estimation tasks, especially in settings like credit-card offers where interpretability, sta-
121 bility, and data limitations matter. Classical econometric models such as Multinomial Logit, Mixed
122 Logit, and Nested or Cross-Nested Logit provide strong theoretical foundations and transparent
123 substitution structures, making them practical and regulator-friendly. Semi-parametric methods like
124 RKHS choice models or Gaussian Process utilities offer also flexible nonlinear modeling without
125 the complexity of deep networks.

126 **4 Testing plans and testing results**

127 To ensure that the implementation is correct and that the reported results are valid, I developed a
128 structured testing pipeline using pytest, consisting of both **unit tests** and **integration tests** for the
129 two model files `model/featureless.py` and `model/featurebased.py` in my provided code.

130 **4.1 Unit Tests**

131 **(a) Tests for `model/featureless.py`**

132 **ExaResBlock**

- 133 • **Test:** `test_exa_res_block_shape`
- 134 • **Checks:**
 - 135 – Given inputs $z_{\text{prev}} \in \mathbb{R}^{B \times d}$ and $e_0 \in \mathbb{R}^{B \times D}$, the output has shape (B, d) .
 - 136 – No NaN values are produced.

137 QuaResBlock

- 138 • **Test:** test_qua_res_block_shape
- 139 • **Checks:**
 - 140 – Given $x \in \mathbb{R}^{B \times D}$, the output shape remains (B, D) .
 - 141 – Output contains no NaN values.

142 (b) Tests for model/featurebased.py

143 make_valid_mask

- 144 • Verifies correct construction of boolean masks for variable-length sequences.

145 masked_softmax

- 146 • Ensures probabilities over valid positions sum to approximately 1.
- 147 • Confirms invalid positions have probability near 0.

148 NonlinearTransformation

- 149 • **Test:** test_nonlinear_transformation_shape
- 150 • **Checks:**
 - 151 – Input $X \in \mathbb{R}^{B \times n \times d}$ produces output of shape (B, n, H, embed) .
 - 152 – No NaN values occur.

153 The unit tests confirm that all core components preserve expected tensor shapes, apply masking
154 logic correctly, and avoid numerical instabilities.

155 4.2 Integration Tests

156 I also implemented end-to-end tests for the two main networks to ensure that the *entire forward pass*
157 behaves correctly.

158 Featureless MainNetwork

- 159 • **Test:** test_main_network_featureless_forward
- 160 • **Checks:**
 - 161 – Output probs and logits have shape (B, D) .
 - 162 – Row-wise probability sums are approximately 1.
 - 163 – Options masked as unavailable ($e_i = 0$) receive near-zero probability.
 - 164 – No NaN values appear.

165 Feature-based MainNetwork

- 166 • **Test:** test_main_network_featurebased_forward
- 167 • **Checks:**
 - 168 – Output log_probs and logits have shape (B, n) .
 - 169 – Probabilities over valid positions (as indicated by lengths) sum to 1.
 - 170 – Probabilities at invalid positions are near zero.
 - 171 – No NaN values occur.

172 These integration tests verify that the complete networks produce well-formed probability distribu-
173 tions, correctly enforce masking, and implement the intended choice-model semantics.

174 **4.3 Execution and Results**

175 All tests were executed using:

176 `pytest -q`

177 All unit and integration tests passed successfully. This demonstrates that:

- 178 • tensor shapes and residual structures are consistent,
- 179 • masking logic is correctly applied,
- 180 • both models generate valid and stable probability distributions,
- 181 • no numerical instabilities (e.g., NaNs) occur.

182 Together, this testing suite provides strong evidence that the implementation is correct at both the
183 component level and system level, and that all experimental results are based on a numerically sound
184 and reliable codebase.

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